Time-block forecast scores: AR(2) versus i.i.d. temporal pooling

Report on Experiments A–C (post `z.init` fix, assertion-guarded)

Score report

17 July 2026 (rev. 18 July 2026: long-memory settings)

Abstract

Two clean experiments compare the AR(2) skew-t model against the i.i.d.-in-time skew-t baseline under the time-block forecast protocol, by CRPS and by the Brier score at the q_{95} threshold, resolved by forecast lead time. **Experiment A** (setting 4, five expanding-window blocks $\times$ 5 datasets): the models tie once the reflected-ridge blocks are excluded; the only visible AR(2) gain sits at lead 1 and is gone by lead 3, exactly as the geometric memory of $\phi = (0.80, -0.35)$ predicts. **Experiment B** (block 1 only, guarded fits, datasets 1–10): at the same setting 4 the tie replicates; at the near-unit-root setting 5, $\phi = (0.15, 0.80)$, the AR(2) advantage becomes real and large — CRPS -31% over leads 1–5 (Wilcoxon $p = 0.040$; sensitivity $p = 0.010$–0.021) and a Brier improvement that is *flat in lead time* (AR(2) better at 14 of 15 leads, mean -0.008), consistent with a memory horizon ($h^* \approx 109$ days) that dwarfs the 15-day window. **Experiment C** (block 1, two long-memory settings, ARFIMA$(0, d, 0)$ with hyperbolic-decay dependence the AR(2) model cannot represent): the advantage survives at the stronger $d = 0.45$ (Hurst 0.95) — CRPS -13%, AR(2) better at *all* 15 Brier leads — but is diluted by a short-window identification failure that flags half the datasets, and vanishes at $d = 0.35$. Across all four settings the advantage tracks the ratio of memory horizon to training-window length, not the Hurst exponent, and remains bounded by the score-ceiling $\Delta\text{CRPS}/\text{CRPS} = 1 - \sqrt{1 - \text{SNR}\,\kappa_z(h)}$. Adding an AR(1) arm (settings 5, 7) decomposes the gain: at the lag-2-dominant setting 5 AR(1) is indistinguishable from i.i.d. and the entire advantage is the *second* lag (AR(2)−AR(1) CRPS -0.75, $W\,p = 0.017$ at leads 1–15), while under long memory (setting 7) both lags contribute and the second-lag increment is the sharpest signal in the study ($W\,p = 0.030$).

1 Data-generating processes and design

All settings share one skeleton and differ *only* in the temporal dependence of the latent processes (`code/analysis/time_block_forecast/simstudy/setup.R`):

component	specification
marginal law	skew-t: $y_{st} = \mathbf{x}'\boldsymbol{\beta} + \lambda z_t + \varepsilon_{st}/\sqrt{\tau_t}$, $\tau \sim$ Gamma$(6,16)$ mixing (t-tails, $\nu \approx 6$), $\lambda = 3$ (right skew), $\beta_0 = 10$
skew latent	$z_t \mid \tau_t \sim \mathrm{HN}(0, \tau_t^{-1/2})$, global ($K = 1$ knot, so the temporal signal lives entirely in τ, z)
spatial	Matérn $\nu = 0.5$ (exponential), range 1, spatial proportion $\gamma = 0.9$; 144 sites uniform on $[0,10]^2$
temporal	Gaussian-copula **AR(2)** (settings 4–5) or **ARFIMA**$(0,d,0)$ (settings 6–7) on $\tau^\star, z^\star$ ($w^\star$), per setting
record	$n_t = 200$ days; expanding-window blocks, each forecasting $H = 15$ days

id	label	temporal structure	persistence
4	strong	AR(2) $\phi = (0.80, -0.35)$	$\rho(F) = 0.592$, $h^* = 6$ d (short memory)
5	near-unit-root	AR(2) $\phi = (0.15, 0.80)$	$\rho(F) = 0.973$, $h^* = 109$ d (short memory)
6	LM $d = 0.35$	ARFIMA$(0, 0.35, 0)$	Hurst 0.85, hyperbolic ACF (long memory)
7	LM $d = 0.45$	ARFIMA$(0, 0.45, 0)$	Hurst 0.95, hyperbolic ACF (long memory)

Settings 4–5 are short-memory ($\sum_h |\rho(h)| < \infty$, ACF $\sim \rho(F)^h$); settings 6–7 are genuinely long-memory ($\sum_h |\rho(h)| = \infty$, ACF $\sim h^{2d-1}$), which no finite-order AR reproduces — so the AR(2) model is *misspecified by construction* there. Experiments A and B use settings 4–5; Experiment C uses 6–7.

Protocol. Fit on a contiguous prefix $y[, 1\!:\!T_o]$ (all sites observed), forecast the window $(T_o, T_o + 15]$ with the seam-conditioned recursion of `forecast_block()`. Scores per lead h: CRPS (`crps_sample`, averaged over sites) and the Brier score for exceedance of the validation block's q_{95}, identical thresholds for both methods, paired by dataset. Both experiments post-date the `z.init` repair; all Experiment-B fits passed the pre-registered consistency guard (truth recovery $\wedge$ predictive spread; see the companion note `tex/guard_effect`), which replaced 13 of 40 attempt-0 chains. Experiment A excludes the four blocks flagged by the same assertions (reflected-ridge fits; `tex/lambda_phiz_ridge`).

2 Experiment A: setting 4, five blocks × five datasets

Expanding-window blocks with seams at $T_o = 50, 80, 110, 140, 170$; 21 clean (dataset, block) pairs.

Reading. The AR(2) gain is visible exactly where geometric memory puts it: lead 1 (-0.308 CRPS, -0.0087 Brier, both the largest gaps in the table) and residually lead 2; from lead 3 onward the gaps oscillate around zero at the noise scale. Pooled over all leads the models tie (paired two-sided tests over the 21 blocks: Brier $p = 0.97$, CRPS $p = 0.78$). With spectral radius 0.592, the seam state is worth $\kappa_z(h) \approx 0.59^h$ of predictable signal: about a third at $h = 1$ and effectively nothing by $h = 5$ — the lead profile of Table 1 is that decay, read through two proper scores.

3 Experiment B: block 1, guarded, datasets 1–10

Block 1 trains on 50 days and forecasts days 51–65. Every kept fit passed the guard; setting-5 datasets 2 and 8 are excluded from the primary analysis (flagged; sensitivity below).

Table 1: Experiment A, setting 4 (strong), clean 21 blocks. Mean score by lead; gap = AR(2) − i.i.d. (negative favours AR(2)).

lead h	CRPS			Brier q_{95}		
	i.i.d.	AR(2)	gap	i.i.d.	AR(2)	gap
1	1.822	1.514	−0.308	0.0277	0.0190	−0.0087
2	2.120	2.048	−0.072	0.0585	0.0552	−0.0034
3	2.273	2.288	+0.015	0.0509	0.0493	−0.0016
4	2.933	2.940	+0.008	0.0644	0.0644	+0.0000
5	3.330	3.318	−0.012	0.0977	0.0989	+0.0012
6	2.785	2.941	+0.156	0.0692	0.0801	+0.0109
7	2.285	2.261	−0.024	0.0518	0.0523	+0.0006
8	2.611	2.576	−0.035	0.0694	0.0701	+0.0007
9	1.968	1.991	+0.023	0.0298	0.0301	+0.0003
10	2.150	2.062	−0.088	0.0679	0.0603	−0.0076
11	2.055	2.085	+0.030	0.0403	0.0417	+0.0014
12	2.372	2.520	+0.148	0.0554	0.0608	+0.0053
13	3.087	3.232	+0.145	0.0712	0.0767	+0.0055
14	2.350	2.267	−0.083	0.0205	0.0231	+0.0026
15	2.432	2.298	−0.134	0.0768	0.0718	−0.0050
1–5	2.495	2.422	−0.074	0.0599	0.0574	−0.0025
1–15	2.438	2.423	−0.015	0.0568	0.0569	+0.0001

4 Experiment C: long-memory settings, block 1, guarded, datasets 1–10

The two long-memory settings replace the AR(2) latent dynamics with ARFIMA$(0, d, 0)$, whose autocorrelation decays hyperbolically ($\rho(h) \sim h^{2d-1}$) rather than geometrically. No finite-order AR reproduces this, so the fitted AR(2) model is misspecified by construction; the question is whether its geometric-memory pooling still beats i.i.d. when the truth has heavier-than-geometric memory. Same block-1 protocol and guard. The short training window interacts badly with long memory: the guard flags 2 of 10 datasets at $d = 0.35$ and 5 of 10 at $d = 0.45$ (both methods, λ stable but under-identified in a 50-day window), so the clean n is smaller here — itself a finding about fitting long memory on short records.

5 Reading the tables

1. The advantage is a dose response — in memory relative to the window, not in the Hurst exponent. Pooled CRPS gap (leads 1–5): −0.058 (setting 4, $\rho(F) = 0.59$), −0.931 (setting 5, $\rho(F) = 0.97$, three-way $n = 7$), −0.022 (setting 6, $d = 0.35$), −0.304 (setting 7, $d = 0.45$). The order is *not* monotone in nominal long-memory strength: the short-memory near-unit-root setting 5 beats both genuinely long-memory settings. What orders the four is the memory horizon measured against the 50-day training window. Setting 5's $h^* \approx 109$ days makes the seam state highly predictive across the whole 15-day window; setting 7's hyperbolic ACF is strong but the AR(2) model can only capture its geometric part, and the 50-day window under-identifies even that (see point 4). Setting 6's short-lag ACF ($\rho(1) \approx 0.4$) sits in the same undetectable band as setting 4. The ceiling $\Delta\mathrm{CRPS}/\mathrm{CRPS} = 1 - \sqrt{1 - \mathrm{SNR}\, \kappa_z(h)}$ still bounds

Table 2: Experiment B, setting 4 (strong), $n = 10$ datasets.

lead h	CRPS			Brier q_{95}		
	i.i.d.	AR(2)	gap	i.i.d.	AR(2)	gap
1	2.550	2.275	−0.275	0.0486	0.0552	+0.0066
2	3.169	3.141	−0.028	0.0923	0.0922	−0.0001
3	1.749	1.766	+0.017	0.0281	0.0236	−0.0045
4	1.861	1.841	−0.020	0.0414	0.0398	−0.0016
5	1.996	2.011	+0.015	0.0524	0.0540	+0.0015
6	2.244	2.259	+0.014	0.0333	0.0351	+0.0019
7	2.628	2.660	+0.032	0.0930	0.0938	+0.0008
8	1.775	1.827	+0.051	0.0358	0.0390	+0.0031
9	1.649	1.685	+0.035	0.0255	0.0262	+0.0007
10	2.172	2.143	−0.029	0.0936	0.0915	−0.0020
11	1.676	1.664	−0.012	0.0333	0.0347	+0.0013
12	2.132	2.096	−0.036	0.0774	0.0762	−0.0011
13	2.589	2.543	−0.046	0.1410	0.1368	−0.0043
14	1.805	1.808	+0.003	0.0311	0.0324	+0.0013
15	2.067	2.058	−0.009	0.0694	0.0682	−0.0013
1–5	2.265	2.207	−0.058	0.0526	0.0530	+0.0004
1–15	2.137	2.118	−0.019	0.0597	0.0599	+0.0002

every case: the seam state is worth only what the *fitted* model's $\kappa_z(h)$ retains of it.

2. The lead profile matches the memory. Setting 4 ($h^* = 6$ d): the gain lives at leads 1–2 and dies by lead 3, in *both* experiments (Tables 1, 2; lead-1 CRPS gaps −0.308, −0.275 — a clean cross-protocol replication). Settings 5 and 7 (long horizon): the gain does *not* decay inside the window — setting 7's Brier gap is negative at *all 15* leads (Table 8) and setting 5's at 11/15, because a long-memory latent state is still informative 15 days out.

3. AR(2)'s advantage over i.i.d. is almost entirely the second lag — and where it is not tells us why. The three-way tables (settings 5, 7) split AR(2)−i.i.d. into AR(1)−i.i.d. (what one lag buys) and AR(2)−AR(1) (the second lag). At the lag-2-dominant setting 5 ($\phi = (0.15, 0.80)$) AR(1) is nearly indistinguishable from i.i.d. (CRPS gap −0.183, leads 1–5) — a single lag cannot represent a lag-2-dominant process — so essentially the whole advantage is the second lag (AR(2)−AR(1) = −0.748; significant at leads 1–15, $W\,p = 0.017$). At the long-memory setting 7 the split is shared: AR(1) captures part of the hyperbolic decay's geometric envelope (CRPS gap −0.129) and the second lag adds a further, cleanly significant increment (AR(2)−AR(1) = −0.175, $W\,p = 0.030$ — the sharpest signal in setting 7). Either way the ordering i.i.d. $\succeq$ AR(1) $\succeq$ AR(2) holds, and the AR(2)−AR(1) gap localizes exactly what the extra lag is worth.

4. CRPS detects what Brier cannot yet. Where the advantage is real (settings 5 and 7) the two scores agree in direction but CRPS reaches significance first: setting 5 CRPS $W\,p = 0.021$–0.040, setting 7 CRPS $W\,p = 0.053$ ($p = 0.077$ on all 10), while both Brier endpoints stay $p > 0.1$. The q_{95} Brier scores one binary tail event per site-day and carries an order of magnitude more sampling noise; at setting 7 its dataset-level gap is also dominated by one dataset ($d = 3$), so with clean $n = 5$ the signed-rank test is mixed even though the lead-averaged gap is negative at

Table 3: Experiment B, setting 5 (near-unit-root): **CRPS** by lead, three-way. Clean $n = 7$ (datasets passing the guard under *all three* methods). Last column = AR(2)−AR(1), the value of the second lag. AR(1) tracks i.i.d. closely; the advantage is almost entirely the second lag.

lead h	i.i.d.	AR(1)	AR(2)	AR(2)−AR(1)
1	2.277	2.060	1.654	−0.406
2	3.736	3.346	2.025	−1.321
3	2.910	2.823	2.316	−0.507
4	3.071	2.946	1.691	−1.255
5	2.717	2.622	2.372	−0.250
6	2.956	2.956	2.166	−0.790
7	2.753	2.708	2.375	−0.333
8	3.123	3.139	2.323	−0.816
9	2.820	2.852	2.765	−0.087
10	4.180	4.187	2.954	−1.233
11	1.836	1.897	2.019	+0.122
12	3.557	3.573	2.587	−0.987
13	1.852	1.884	2.029	+0.145
14	3.019	3.040	2.571	−0.469
15	1.975	2.003	2.155	+0.151
1–5	2.942	2.759	2.011	−0.748
1–15	2.852	2.802	2.267	−0.536

every lead. Powering Brier at these gaps needs $\sim$30–50 clean datasets (`run_block1_guarded.R --datasets=11:50`).

5. Long memory on a short window is an identification problem, not just a modelling one. Setting 7 flags half its datasets: a 50-day window can realise a nearly flat draw of a long-memory z process, which under-identifies λ (stable but too small), and the AR(2) misspecification compounds it. This is why setting 7, despite the strongest nominal memory, does not beat setting 5: the advantage it *could* deliver is partly eaten by fitting difficulty on 50 days. The direct test of "does long memory need a long window" is therefore block 5 (train 170 days), not more datasets — the natural next experiment.

6. What the ties at settings 4 and 6 mean. Neither is evidence that AR(2) pooling is useless; each says the process memory is short relative to the 15-day window, so most scored leads lie beyond it and any pooled endpoint dilutes its own signal. The informative settings are the ones whose horizon exceeds the window (5 and 7), which is where the advantage appears.

Provenance. Experiment A: `simstudy/results/4-*.RData` (post-fix rerun), scored by `scores.R`; clean-block set from `fit_diag_4.RData`. Experiments B and C: `block1_positive_control/results/blk1-*.RDa` (guarded driver), scored by the same protocol; flagged datasets from `blk1_diag` tables. Guard counterfactual and the score-channel decomposition of the reflected ridge: companion notes `tex/guard_effect` and `tex/z_init_bug`.

Table 4: Experiment B, setting 5 (near-unit-root): **Brier** q_{95} by lead, three-way. Clean $n = 7$.

lead h	i.i.d.	AR(1)	AR(2)	AR(2)−AR(1)
1	0.0489	0.0459	0.0490	+0.0031
2	0.0601	0.0575	0.0576	+0.0001
3	0.0698	0.0715	0.0805	+0.0090
4	0.0173	0.0173	0.0178	+0.0005
5	0.0974	0.0961	0.0978	+0.0018
6	0.0206	0.0206	0.0192	−0.0014
7	0.0686	0.0691	0.0607	−0.0084
8	0.0404	0.0416	0.0390	−0.0026
9	0.0795	0.0823	0.0771	−0.0051
10	0.0919	0.0931	0.0807	−0.0123
11	0.0119	0.0138	0.0097	−0.0042
12	0.0650	0.0660	0.0575	−0.0085
13	0.0250	0.0269	0.0232	−0.0037
14	0.0431	0.0439	0.0408	−0.0030
15	0.0171	0.0188	0.0153	−0.0035
1–5	0.0587	0.0577	0.0606	+0.0029
1–15	0.0504	0.0509	0.0484	−0.0025

Table 5: Experiment B paired one-sided CRPS tests (H_1: first method better than second), per-dataset units. Setting 5 is three-way (clean $n = 7$); AR(2)−AR(1) is the value of the second lag. Brier gaps (all $p > 0.1$) are in the per-lead tables.

setting	contrast (CRPS)	gap	t p	Wilcoxon p
4	AR(2) − i.i.d., leads 1–5	−0.058	0.135	0.154
4	AR(2) − i.i.d., 1–15	−0.019	0.216	0.270
5	AR(1) − i.i.d., leads 1–5	−0.183	0.091	0.136
5	AR(2) − i.i.d., leads 1–5	−0.931	0.075	0.054
5	**AR(2) − AR(1)**, leads 1–5	−0.748	0.106	0.075
5	**AR(2) − AR(1)**, 1–15	−0.536	0.110	**0.017**

Table 6: Experiment C, setting 6 (LM $d = 0.35$, Hurst 0.85), clean $n = 8$.

lead h	CRPS			Brier q_{95}		
	i.i.d.	AR(2)	gap	i.i.d.	AR(2)	gap
1	1.788	1.575	−0.212	0.0089	0.0214	+0.0125
2	2.227	2.196	−0.032	0.0383	0.0447	+0.0063
3	1.521	1.742	+0.221	0.0085	0.0128	+0.0043
4	2.412	2.540	+0.128	0.0673	0.0691	+0.0018
5	2.386	2.172	−0.214	0.0296	0.0272	−0.0024
6	2.758	2.696	−0.062	0.0536	0.0548	+0.0012
7	1.881	1.840	−0.041	0.0122	0.0144	+0.0022
8	2.942	2.780	−0.161	0.0764	0.0740	−0.0024
9	2.546	2.540	−0.006	0.0591	0.0617	+0.0025
10	1.732	1.770	+0.038	0.0110	0.0126	+0.0016
11	2.938	2.804	−0.134	0.0664	0.0629	−0.0034
12	2.989	3.030	+0.041	0.0968	0.0981	+0.0014
13	3.232	3.227	−0.005	0.0576	0.0589	+0.0013
14	3.344	3.239	−0.105	0.1035	0.1008	−0.0027
15	2.737	2.751	+0.014	0.0796	0.0829	+0.0033
1–5	2.067	2.045	−0.022	0.0305	0.0350	+0.0045
1–15	2.496	2.460	−0.035	0.0513	0.0531	+0.0018

Table 7: Experiment C, setting 7 (LM $d = 0.45$, Hurst 0.95): **CRPS** by lead, three-way. Clean $n = 5$. Last column = AR(2)−AR(1). Here AR(1) captures part of the geometric decay and the second lag adds the rest.

lead h	i.i.d.	AR(1)	AR(2)	AR(2)−AR(1)
1	1.750	1.462	1.633	+0.172
2	2.257	1.931	1.960	+0.029
3	2.223	1.995	1.659	−0.337
4	2.013	1.981	1.738	−0.244
5	3.119	3.350	2.856	−0.494
6	2.640	2.568	2.213	−0.355
7	2.287	2.357	1.946	−0.412
8	4.400	4.151	3.527	−0.624
9	2.692	2.889	2.479	−0.410
10	2.970	3.299	2.957	−0.342
11	3.137	3.460	3.178	−0.282
12	2.731	2.969	2.705	−0.264
13	2.467	2.689	2.447	−0.242
14	2.487	2.689	2.451	−0.238
15	3.249	3.612	3.343	−0.269
1–5	2.273	2.144	1.969	−0.175
1–15	2.695	2.760	2.473	−0.287

Table 8: Experiment C, setting 7 (LM $d = 0.45$, Hurst 0.95): **Brier** q_{95} by lead, three-way. Clean $n = 5$. The monotone ordering i.i.d. > AR(1) > AR(2) holds at nearly every lead.

lead h	i.i.d.	AR(1)	AR(2)	AR(2)−AR(1)
1	0.0560	0.0427	0.0394	−0.0033
2	0.0459	0.0329	0.0229	−0.0100
3	0.0552	0.0434	0.0302	−0.0132
4	0.0505	0.0416	0.0319	−0.0098
5	0.0586	0.0511	0.0394	−0.0116
6	0.0508	0.0486	0.0490	+0.0004
7	0.0442	0.0345	0.0262	−0.0083
8	0.1744	0.1649	0.1494	−0.0154
9	0.0652	0.0554	0.0440	−0.0114
10	0.1085	0.1033	0.0914	−0.0119
11	0.1056	0.0978	0.0876	−0.0102
12	0.0984	0.0937	0.0832	−0.0105
13	0.0481	0.0411	0.0344	−0.0067
14	0.0584	0.0485	0.0406	−0.0079
15	0.1073	0.1002	0.0938	−0.0064
1–5	0.0532	0.0424	0.0328	−0.0096
1–15	0.0751	0.0666	0.0576	−0.0091

Table 9: Experiment C paired one-sided CRPS tests (H_1: first method better than second). Setting 7 is three-way (clean $n = 5$); AR(2)−AR(1) is the value of the second lag.

setting	contrast (CRPS)	gap	t p	Wilcoxon p
6	AR(2) − i.i.d., leads 1–5	−0.022	0.336	0.417
7	AR(1) − i.i.d., leads 1–5	−0.129	0.146	0.209
7	AR(2) − i.i.d., leads 1–5	−0.304	0.067	0.053
7	**AR(2) − AR(1)**, leads 1–5	−0.175	**0.030**	**0.030**
7	**AR(2) − AR(1)**, 1–15	−0.287	0.125	0.140